

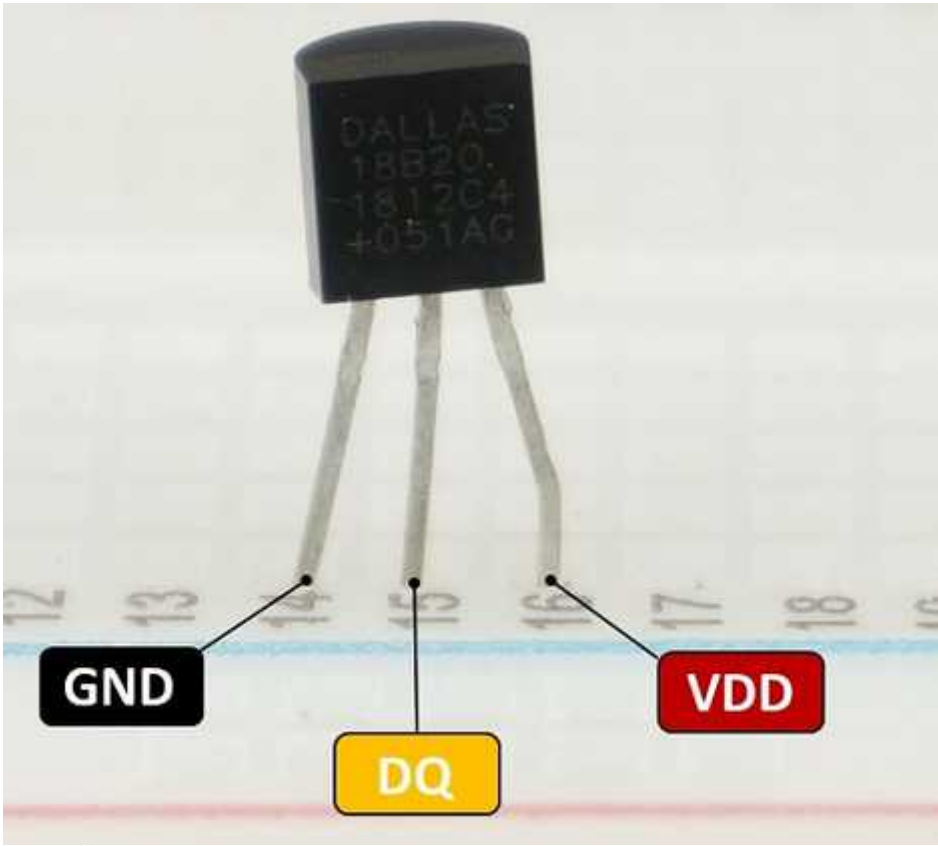
Guide for DS18B20 Temperature Sensor with Arduino

This guide shows how to use the DS18B20 temperature sensor with the Arduino board. You'll learn how to wire the sensor, install the required libraries and get temperature from one or multiple DS18B20 sensors.

Introducing DS18B20 Temperature Sensor

The [DS18B20 temperature sensor](#) is a one-wire digital temperature sensor. This means that it just requires one data line (and GND) to communicate with the Arduino.

It can be powered by an external power supply or it can derive power from the data line (called “parasite mode”), which eliminates the need for an external power supply.



DS18B20	Arduino
GND	GND
DQ	Any digital pin (with 4.7k Ohm pull-up resistor)
VDD	5V (normal mode) or GND (parasite mode)

Each DS18B20 temperature sensor has a unique 64-bit serial code. This allows you to wire multiple sensors to the same data wire. So, you can get temperature from multiple sensors using just one Arduino digital pin.

The DS18B20 temperature sensor is also available in [waterproof version](#).



Here's a summary of the most relevant specs of the DS18B20 temperature sensor:

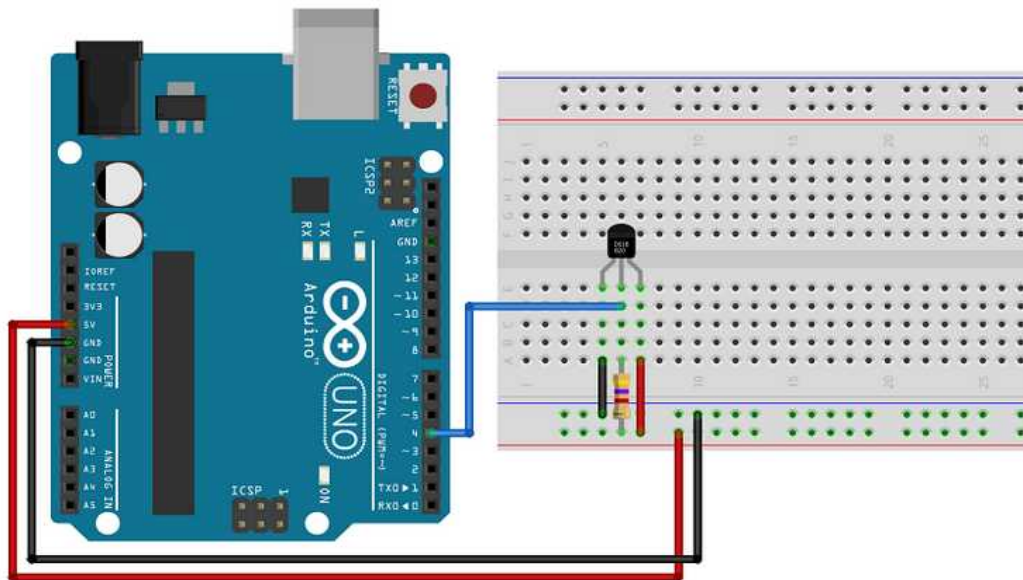
- Communicates over one-wire bus communication
- Power supply range: 3.0V to 5.5V
- Operating temperature range: -55°C to +125°C
- Accuracy +/-0.5 °C (between the range -10°C to 85°C)

For more information consult the [DS18B20 datasheet](#).

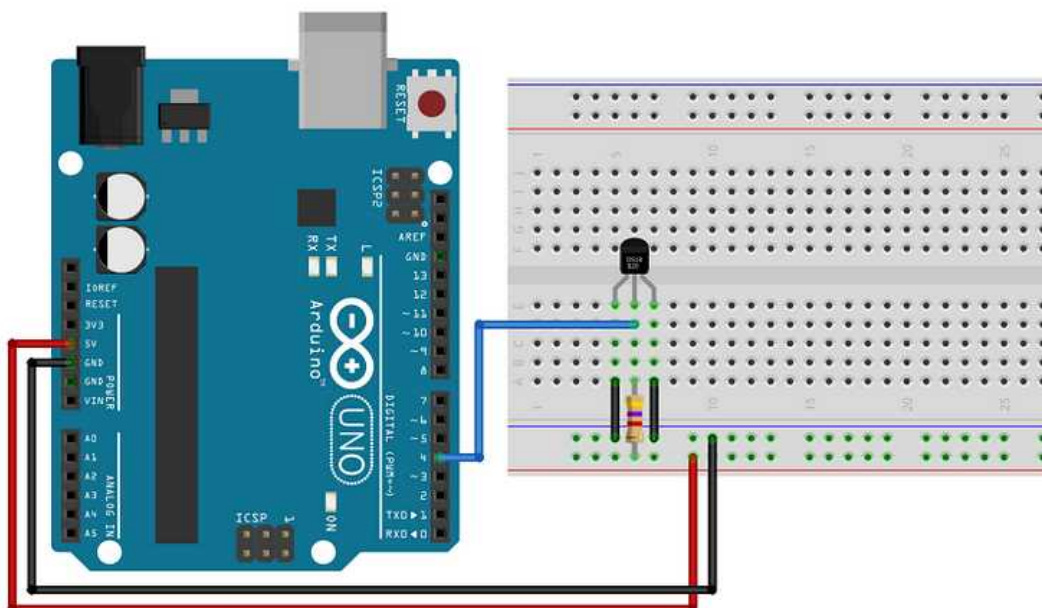
Schematic

The sensor can operate in two modes:

- **Normal mode:** 3-wire connection is needed. You provide power to the VDD pin. Here's the schematic you need to follow:



- **Parasite mode:** You only need data and GND. The sensor derives its power from the data line. In this case, here's the schematic you need to follow:



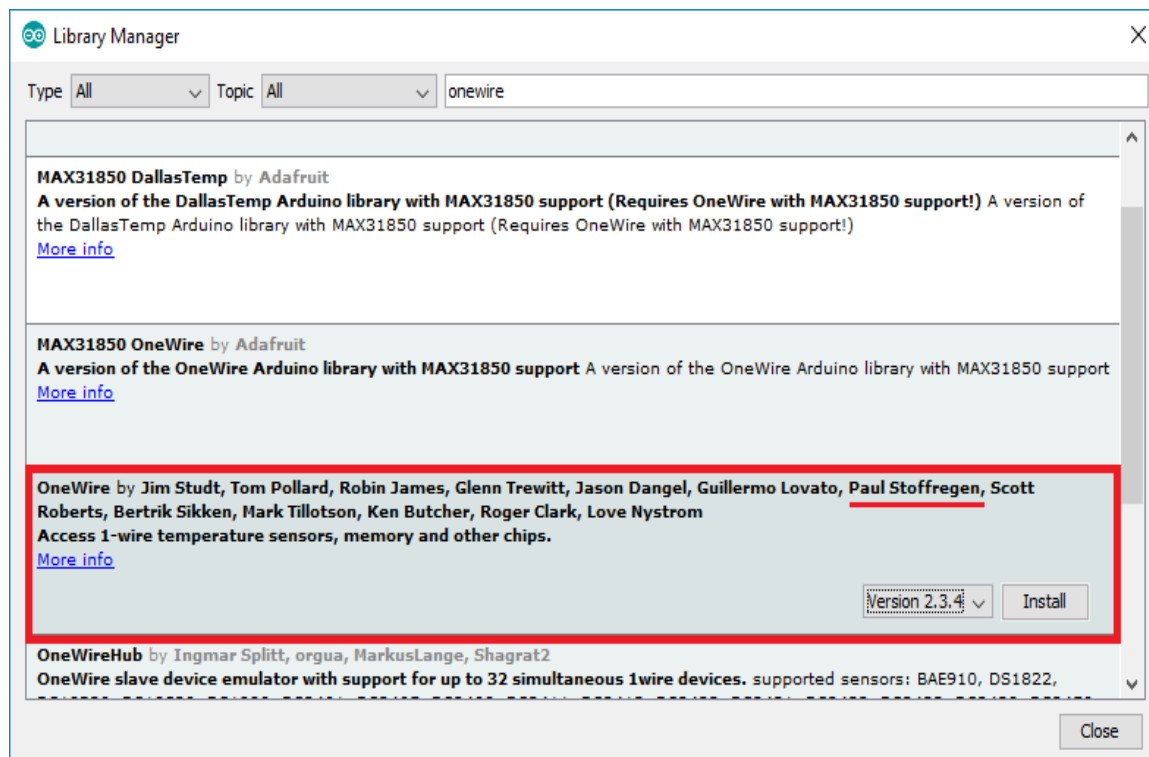
You can read the temperature of more than one sensor at the same time using just one Arduino digital pin. For that, you just need to wire together all the sensors data pins to an Arduino digital pin.

Upload Code – Single DS18B20

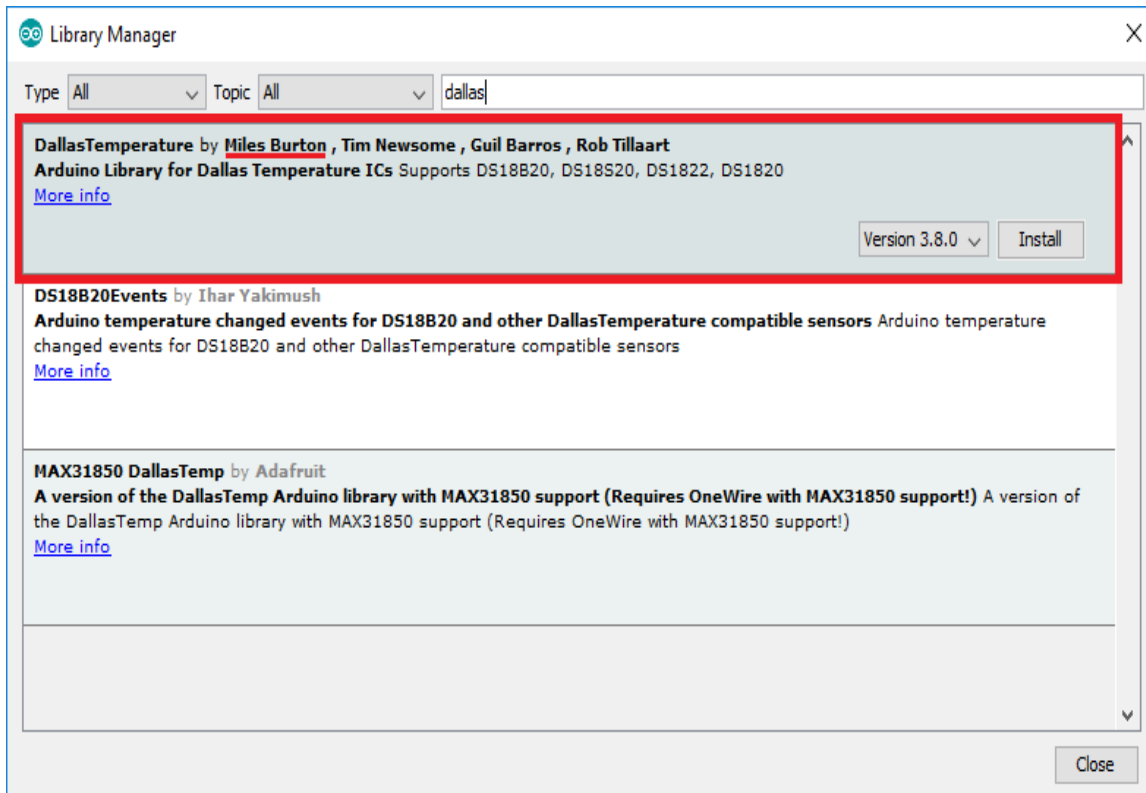
To interface with the DS18B20 temperature sensor, you need to install the [One Wire library by Paul Stoffregen](#) and the [Dallas Temperature library](#). Follow the next steps to install those libraries.

Installing Libraries

1. Open your Arduino IDE and go to **Sketch > Include Library > Manage Libraries**. The Library Manager should open.
2. Type “**OneWire**” in the search box and install the OneWire library by Paul Stoffregen.



3. Then, search for “**Dallas**” and install the Dallas Temperature library by Miles Burton.



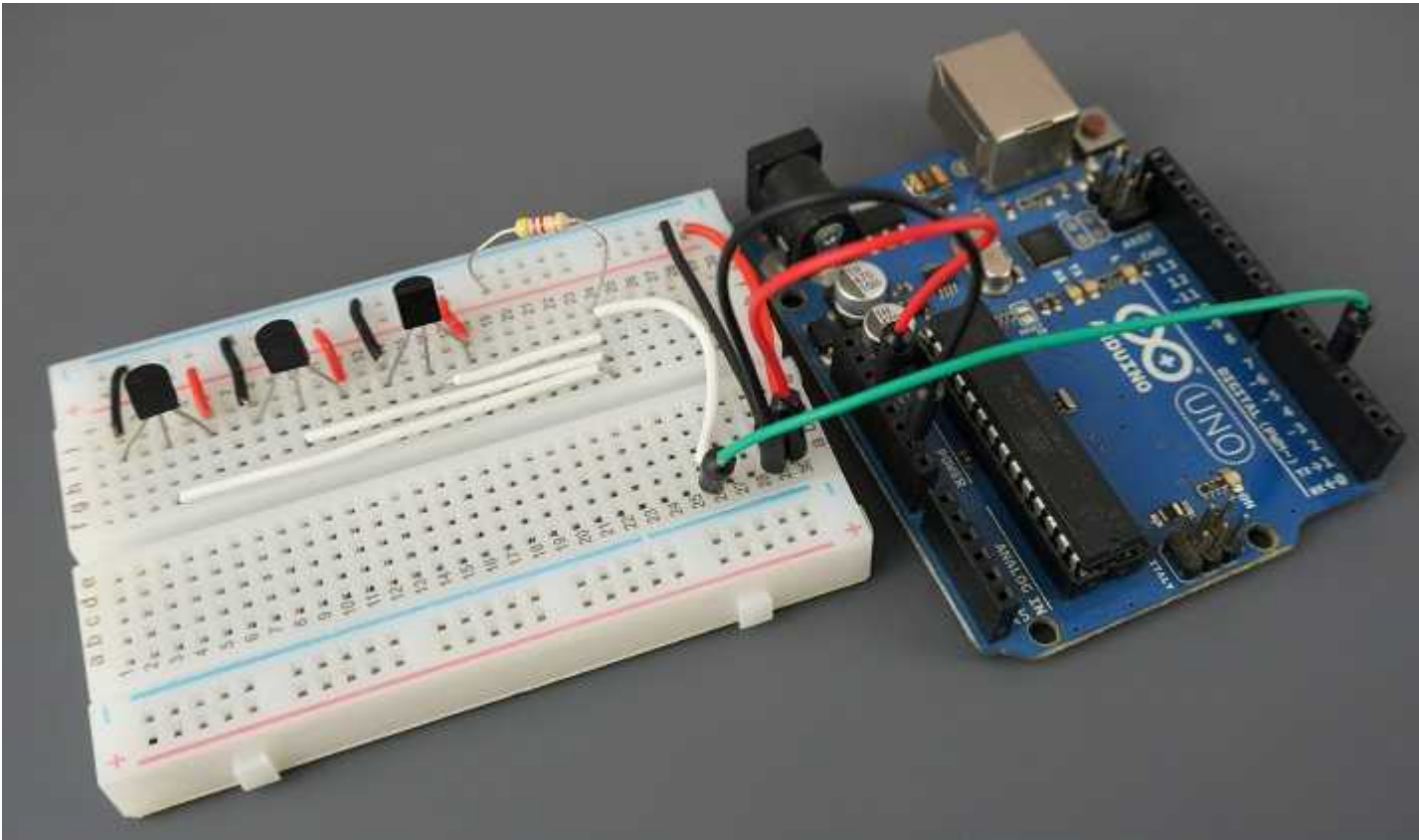
After installing the needed libraries, upload the following code to your Arduino board. This sketch is based on an example from the Dallas Temperature library.

```
/*  
  Rui Santos  
  Complete project details at https://randomnerdtutorials.com  
  Based on the Dallas Temperature Library example  
  */  
  
#include <OneWire.h>  
#include <DallasTemperature.h>  
  
// Data wire is connctec to the Arduino digital pin 4  
#define ONE_WIRE_BUS 4  
  
// Setup a oneWire instance to communicate with any OneWire devices  
OneWire oneWire(ONE_WIRE_BUS);  
  
// Pass our oneWire reference to Dallas Temperature sensor  
DallasTemperature sensors(&oneWire);  
  
void setup(void)  
{  
  // Start serial communication for debugging purposes  
  Serial.begin(9600);  
  // Start up the library  
  sensors.begin();  
}  
  
void loop(void){  
  // Call sensors.requestTemperatures() to issue a global temperature and Requests to all devices on  
  the bus  
  sensors.requestTemperatures();  
  
  Serial.print("Celsius temperature: ");  
  // Why "byIndex"? You can have more than one IC on the same bus. 0 refers to the first IC on the  
  wire  
  Serial.print(sensors.getTempCByIndex(0));  
  Serial.print(" - Fahrenheit temperature: ");  
  Serial.println(sensors.getTempFByIndex(0));  
  delay(5000);  
}
```

The `getTempCByIndex()` and the `getTempFByIndex()` methods accept the index of the temperature sensor. Because we're using just one sensor its index is 0. If you have more than one sensor, you use index 0 for the first sensor, index 1 for the second sensor, and so on.

There are many different ways to get the temperature from DS18B20 temperature sensors. If you're using just one single sensor, this is one of the easiest and simplest ways.

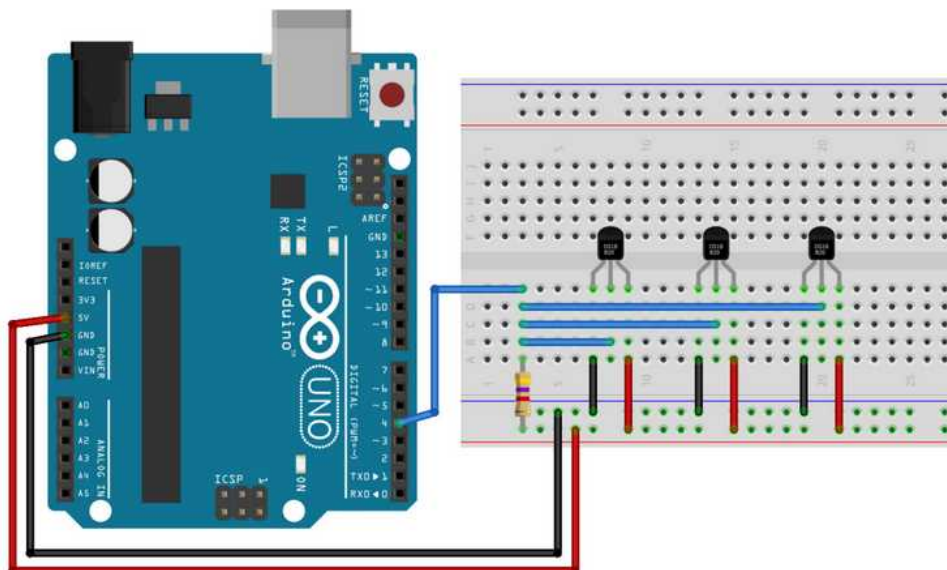
Getting Temperature from Multiple DS18B20 Sensors



The DS18B20 temperature sensor communicates using one-wire protocol and each sensor has a unique 64-bit serial code, so you can read the temperature from multiple sensors using just one single Arduino digital Pin.

Schematic

To read the temperature from multiple sensors, you just need to wire all data lines together as shown in the following schematic diagram:



Upload Code – Multiple DS18B20

Then, upload the following code. It scans for all devices on Pin 4 and prints the temperature for each one. This sketch is based on the example provided by the DallasTemperature library.

```
/*
 * Rui Santos
 * Complete Project Details http://randomnerdtutorials.com
 */

#include <OneWire.h>
#include <DallasTemperature.h>

// Data wire is plugged into port 4 on the Arduino
#define ONE_WIRE_BUS 4
// Setup a oneWire instance to communicate with any OneWire devices (not just Maxim/Dallas temperature ICs)
OneWire oneWire(ONE_WIRE_BUS);

// Pass our oneWire reference to Dallas Temperature.
DallasTemperature sensors(&oneWire);

int numberOfDevices; // Number of temperature devices found

DeviceAddress tempDeviceAddress; // We'll use this variable to store a found device address

void setup(void) {
  // start serial port
  Serial.begin(9600);

  // Start up the library
  sensors.begin();

  // Grab a count of devices on the wire
  numberOfDevices = sensors.getDeviceCount();

  // locate devices on the bus
  Serial.print("Locating devices...");
  Serial.print("Found ");
  Serial.print(numberOfDevices, DEC);
  Serial.println(" devices.");

  // Loop through each device, print out address
  for(int i=0;i<numberOfDevices; i++) {
    // Search the wire for address
    if(sensors.getAddress(tempDeviceAddress, i)) {
      Serial.print("Found device ");
      Serial.print(i, DEC);
      Serial.print(" with address: ");
      printAddress(tempDeviceAddress);
      Serial.println();
    } else {
      Serial.print("Found ghost device at ");
      Serial.print(i, DEC);
      Serial.print(" but could not detect address. Check power and cabling");
    }
  }
}
```



```

    }
}

void loop(void) {
    sensors.requestTemperatures(); // Send the command to get temperatures

    // Loop through each device, print out temperature data
    for(int i=0;i<numberOfDevices; i++) {
        // Search the wire for address
        if(sensors.getAddress(tempDeviceAddress, i)){

            // Output the device ID
            Serial.print("Temperature for device: ");
            Serial.println(i,DEC);

            // Print the data
            float tempC = sensors.getTempC(tempDeviceAddress);
            Serial.print("Temp C: ");
            Serial.print(tempC);
            Serial.print(" Temp F: ");
            Serial.println(DallasTemperature::toFahrenheit(tempC)); // Converts tempC to
Fahrenheit
        }
    }
    delay(5000);
}

// function to print a device address
void printAddress(DeviceAddress deviceAddress) {
    for (uint8_t i = 0; i < 8; i++) {
        if (deviceAddress[i] < 16) Serial.print("0");
        Serial.print(deviceAddress[i], HEX);
    }
}

```

Source:

<https://randomnerdtutorials.com/guide-for-ds18b20-temperature-sensor-with-arduino/pa>